LOVELY PROFESSIONAL UNIVERSITY

MITTAL SCHOOL OF BUSINESS

Project Report

Predictive Analytics

INTM591

Exploratory Data Analysis with Python and IBM SPSS Modeler

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Project Definition

Conduct Exploratory Data Analysis (EDA) on the "myntra_dataset_ByScraping.csv" dataset (men's jeans products obtained from scraping Myntra). This involves understanding the data, dealing with missing values and duplicates, statistical analysis, data visualization to discover patterns and relationships, finding correlation, and obtaining valuable business insights through Python and IBM SPSS Modeler.

Outcomes/Learning

- Developed skills in data cleaning, descriptive analysis, visualization, and insights in Python; acquired knowledge about drag-and-drop data mining, auditing, and graph building with SPSS Modeler.
- Gained skills in recognizing data quality problems (data duplication, discrepancies between discount prices) and turning raw data from e-commerce into insights.

Required Tool

- **Python** (JUPYTER-Notebook / Google Collab) — pandas, matplotlib, seaborn.
- **IBM SPSS Modeler** — for visual data flows, Data Audit, graphs, and statistics nodes.

Working

- Firstly, we can download the "Myntra Sales" dataset from the Kaggle for the Exploratory Data Analysis.
- Then that "Myntra Sales" dataset csv file we can loaded into both the tools.
- After loading the dataset, we can perform the data audit, cleaning and preprocessing for making the data more reliable and accurate for analysis.
- We can perform the Exploratory Data Analysis on the dataset in order to generated the summary statistics. Make Distribution of the data, find the correlation between the data and calculate brand wise aggregations in order to get the more meaningful insights from the dataset.
- For making analysis makes more easier and effective. We can created the visualizations to uncover the patterns, brand performance and customer rating trends.

Step 1: Data Loading and Initial Understanding (Python)

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Code JupyterLab Python 3 (ipykernel)

1. Data Loading and Intial Understanding

Importing the Libraries

[2]:
```
# Import the Libraries for Analysis

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Dataset

[3]:
```
# Load the Myntra Sales Dataset for into dataframe
df = pd.read_csv('myntra_dataset_ByScraping.csv')
df.head()
```

[3]:

	brand_name	pants_description	price	MRP	discount_percent	ratings	number_of_ratings
0	WROGN	Men Loose Fit Cotton Jeans	1374.0	2499.0	0.45	4.2	57.0
1	Flying Machine	Men Slim Fit Jeans	1829.0	2999.0	0.39	4.6	5.0
2	Roadster	Men Pure Cotton Jeans	974.0	2499.0	0.61	3.6	1100.0
3	Bene Kleed	Relaxed Fit Denim Jeans	873.0	2299.0	0.62	4.0	4800.0
4	Levis	Men 511 Slim Fit Jeans	1478.0	2899.0	0.49	4.3	264.0

Data Structure of Myntra Sales Dataset

[4]:
```
# Dimensions of the "Myntra Sales" Dataset
df.shape
```

[4]:
```
(52120, 7)
```

[5]:
```
# Brief information of the "Myntra Sales" Dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 52120 entries, 0 to 52119
Data columns (total 7 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   brand_name          52120 non-null  object
 1   pants_description   52120 non-null  object
 2   price               52120 non-null  float64
 3   MRP                 52120 non-null  float64
 4   discount_percent    52120 non-null  float64
 5   ratings             52120 non-null  float64
 6   number_of_ratings   52120 non-null  float64
dtypes: float64(5), object(2)
memory usage: 2.8+ MB
```

[7]:
```
# Handling the missing data
df.isnull().sum()
```

[7]:
```
brand_name           0
pants_description    0
price                0
MRP                  0
discount_percent     0
ratings              0
number_of_ratings    0
dtype: int64
```

Key Observations

- In Jupyter Notebook, the dataset was loaded using pd.read_csv(). For, loading the "Myntra Sales" dataset we can import the library of 'Pandas'.
- The data set consists of 52,120 rows and 7 columns (brand name, pant description, price, MRP, discount percentage, ratings, number of ratings).
- There are no missing values in the data set. This means the complete dataset is maintained.
- Data types were verified and basic info was displayed.

Step 2: Data Quality Check & Cleaning (Python)

```
[8]: # Check the duplicates values from the dataset
     df.duplicated()

[8]: 0        False
     1        False
     2        False
     3        False
     4        False
              ...
     52115     True
     52116     True
     52117     True
     52118     True
     52119     True
     Length: 52120, dtype: bool
```

```
[11]: Discount_Percent = df['discount_percent'].clip(upper= 1.0)
      Discount_Percent.head()

[11]: 0    0.45
      1    0.39
      2    0.61
      3    0.62
      4    0.49
      Name: discount_percent, dtype: float64

[12]: df.head()
```

[12]:

	brand_name	pants_description	price	MRP	discount_percent	ratings	number_of_ratings
0	WROGN	Men Loose Fit Cotton Jeans	1374.0	2499.0	0.45	4.2	57.0
1	Flying Machine	Men Slim Fit Jeans	1829.0	2999.0	0.39	4.6	5.0
2	Roadster	Men Pure Cotton Jeans	974.0	2499.0	0.61	3.6	1100.0
3	Bene Kleed	Relaxed Fit Denim Jeans	873.0	2299.0	0.62	4.0	4800.0
4	Levis	Men 511 Slim Fit Jeans	1478.0	2899.0	0.49	4.3	264.0

```
[9]: # Remove the duplicate values from the dataset
df.drop_duplicates().reset_index(drop=True)
```

[9]:

	brand_name	pants_description	price	MRP	discount_percent	ratings	number_of_ratings
0	WROGN	Men Loose Fit Cotton Jeans	1374.0	2499.0	0.45	4.2	57.0
1	Flying Machine	Men Slim Fit Jeans	1829.0	2999.0	0.39	4.6	5.0
2	Roadster	Men Pure Cotton Jeans	974.0	2499.0	0.61	3.6	1100.0
3	Bene Kleed	Relaxed Fit Denim Jeans	873.0	2299.0	0.62	4.0	4800.0
4	Levis	Men 511 Slim Fit Jeans	1478.0	2899.0	0.49	4.3	264.0
...	...	...	...	...	...	...	...
35068	United Colors of Benetton	Men Slim Fit Mid-Rise Jeans	949.0	3049.0	21.00	3.8	6.0
35069	HERE&NOW	Men Slim Fit Jeans	759.0	1899.0	0.60	4.0	63.0
35070	Pepe Jeans	Men Vapour Slim Fit Jeans	759.0	1899.0	0.60	4.0	63.0
35071	Pepe Jeans	Men Slim Fit Jeans	759.0	1899.0	0.60	4.0	63.0
35072	Celio	Men Straight Fit Jeans	759.0	1899.0	0.60	4.0	63.0

35073 rows × 7 columns

Key Observations

- Duplicate records check (detected 17,047 duplicates – probably caused by web scraping).
- The duplicates were eliminated using df.drop_duplicates().
- Discrepancies in discount_percent field values (some records had discount greater than 1, whereas others were between 0 and 1) were observed.
- There are removing the duplicated data from the 35073 rows and 7 columns.
- After cleaning and preprocessing the data, it becomes more reliable and accurate for analysis.
- The data becomes more reliable and accurate it makes the predictions more accurate.

Step 3: Descriptive Statistics & Basic Insights (Python)

```
[6]: # Describe the data using descriptive Statistics
df.describe()
```

[6]:

	price	MRP	discount_percent	ratings	number_of_ratings
count	52120.000000	52120.000000	52120.000000	52120.000000	52120.000000
mean	1594.515445	3180.398438	1.648256	3.997794	91.568937
std	1495.972325	2201.883218	4.687529	0.420404	433.918513
min	337.000000	499.000000	0.020000	1.000000	5.000000
25%	989.000000	2499.000000	0.400000	3.800000	16.000000
50%	1439.000000	2999.000000	0.500000	4.000000	35.000000
75%	1829.000000	3499.000000	0.630000	4.200000	74.000000
max	54000.000000	72000.000000	64.000000	5.000000	30700.000000

Key Observations

- We can used df.describe() and group-by aggregation methods were used to draw conclusions.
- The Mean price of products ₹1,594.52 and Mean MRP of the products ₹3,180.40.
- The Mean discount on the products is 1.65% (some outliers may need data cleaning).
- The Ratings distribution is between the 1.0 & 5.0 (average rating: 3.99)
- The 10 most active brands in terms of products sold: WROGN (5,303) and Flying Machine (5,197), followed by United Colors of Benetton, Roadster, etc.
- Brand-wise statistics (top brands by total ratings received) Roadster leads in total customer engagement (653,454 ratings) despite lower average price.

Step 4: Exploratory Visualizations & Pattern Analysis (Python)

Univariate Analysis:

Univariate analysis refers to the process of analyzing a particular variable within a data set and understanding its attributes like distribution, central tendency, dispersion, and whether it has any outliers.

Univariate analysis is implemented through packages such as Pandas, NumPy, Matplotlib, and Seaborn in Python.

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3. Exploratory Visualizations and Pattern Analysis

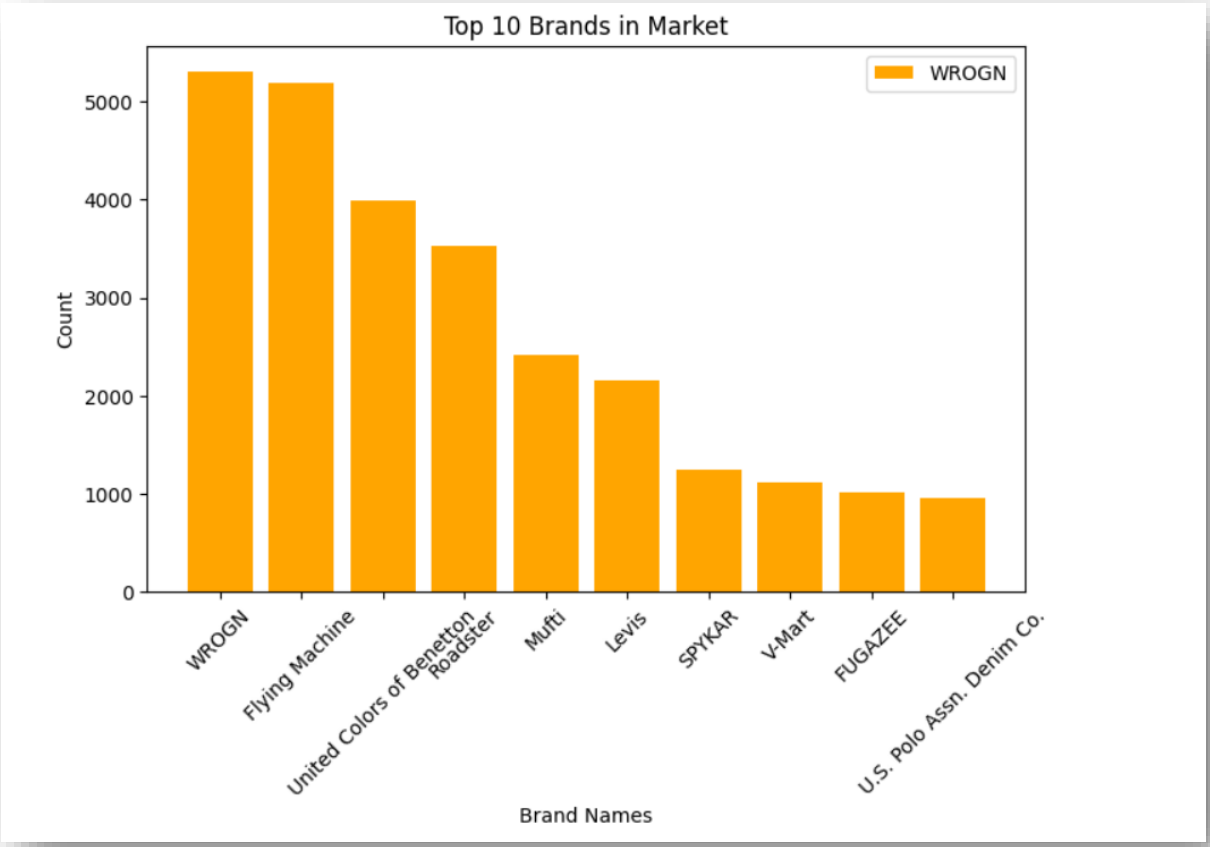
Univariate Analysis

[13]:
```
# Check the Top 10 Competitors of Myntra
df['brand_name'].value_counts()[:10]
```

[13]:
```
brand_name
WROGN                        5303
Flying Machine               5197
United Colors of Benetton    3985
Roadster                     3523
Mufti                        2416
Levis                        2153
SPYKAR                       1247
V-Mart                       1118
FUGAZEE                      1019
U.S. Polo Assn. Denim Co.     953
Name: count, dtype: int64
```

[25]:
```
# Create a Visualization of Univariate Analysis of the Top 10 Brand Names
brands = df['brand_name'].value_counts().reset_index().sort_values(by = 'count', ascending = False)[:10]
plt.figure(figsize=(8,6))
plt.bar(brands['brand_name'], brands['count'], color='orange')
plt.xlabel('Brand Names')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.title('Top 10 Brands in Market')
plt.legend(df['brand_name'])
plt.show()
```

Bar Chart of Top 10 Brands in Market



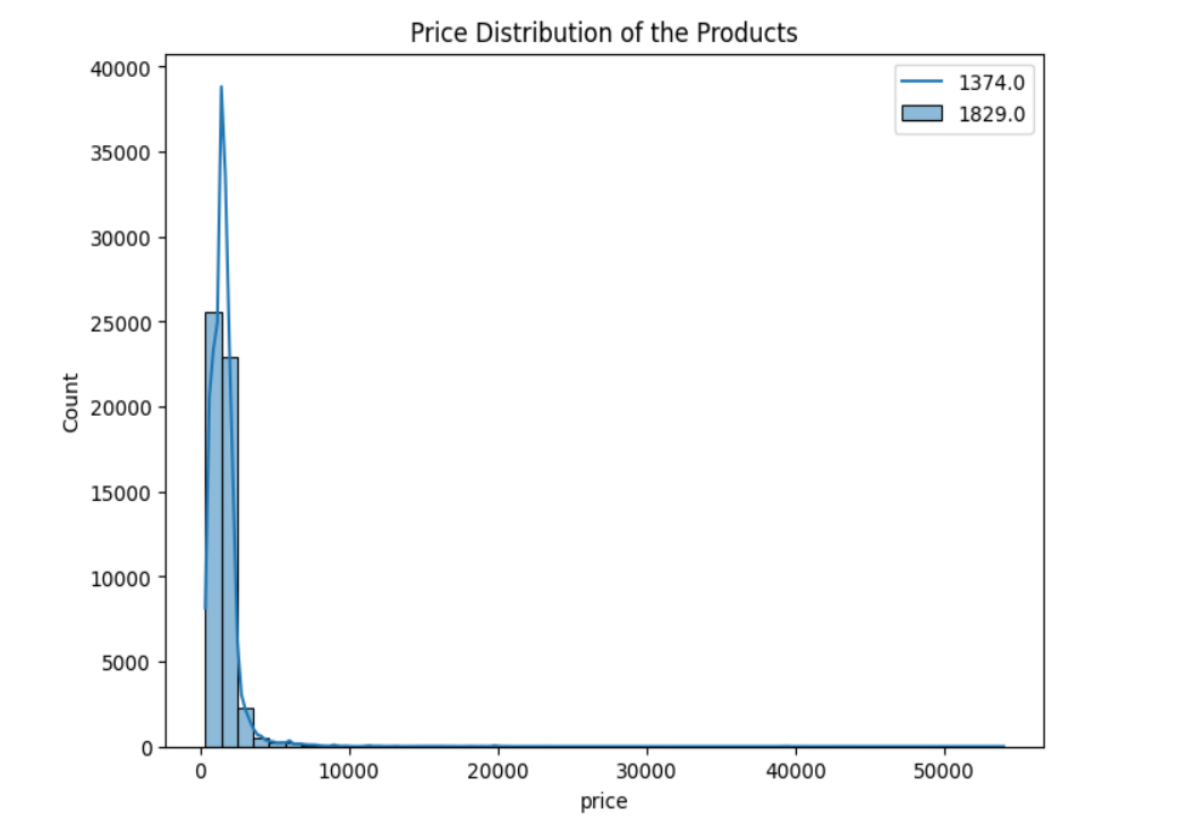
Visualization Findings

- Brand WROGN has the highest count.
- Brand Flying Machine is just behind the leading brand.
- Brand United Colors of Benetton and Brand Roadster have moderate presence.
- There is a consistent decrease in the number of counts of brands starting from the lower ranks.
- Brands Mufti and Levis have mid-level counts.
- Brands SPYKAR, V-Mart, and FUGAZEE have the lowest counts.
- Brand U.S. Polo Assn. Denim Co. has the lowest presence in the given dataset.
- The distribution is positively skewed to the right.
- Some brands have high counts whereas others have lower counts.

Price Distribution of the Products

```
[15]: # Average Selling Price of the Products "Pants and Trousers"
      Avg_Price = df['price'].mean()
      print(Avg_Price)

      1594.5154451266308

[24]: # Price Distribution of the Products
      plt.figure(figsize=(8,6))
      sns.histplot(df['price'], bins = 50, kde=True)
      plt.title("Price Distribution of the Products")
      plt.legend(df['price'])
      plt.show()
```

Histogram of Price Distribution of the Products



Visualization Findings

- Most of the products have their prices clustered around the lower end of the price spectrum (₹1000-₹3000).
- There is a strong right skewness in the pricing of the products.
- A few products have prices that are abnormally high.
- The products are mostly budget to mid-price range products and not premium-priced products.
- There is a steep peak in the price distribution graph for prices between ₹1300-1800.
- Products whose prices are more than ₹10,000 are very few in numbers.
- As the price increases, the number of products decreases.

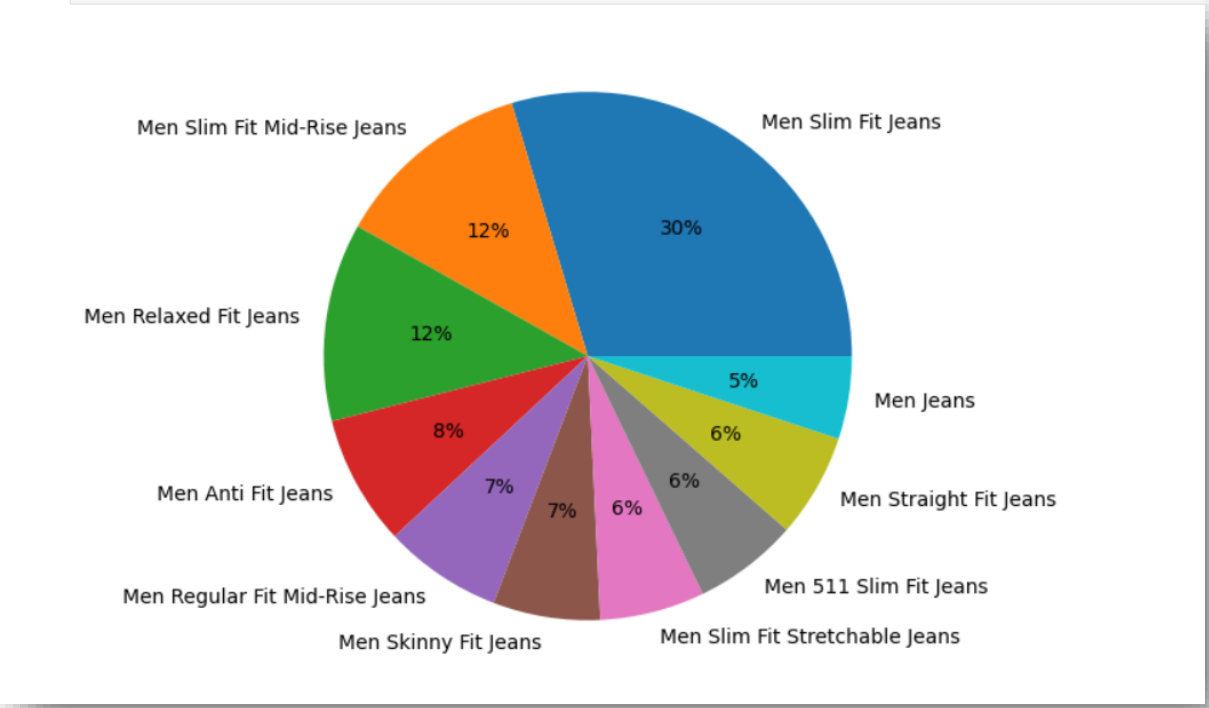
Top 10 Products Purchase By Consumers

[20]:
```
# Top 10 Products "Pants and Trousers" Purchase by Consumers
Pants = df['pants_description'].value_counts()[:10].reset_index().sort_values(by='count', ascending=False)
print(Pants)
```

```
                pants_description  count
0              Men Slim Fit Jeans   7850
1     Men Slim Fit Mid-Rise Jeans   3223
2           Men Relaxed Fit Jeans   3212
3              Men Anti Fit Jeans   2112
4  Men Regular Fit Mid-Rise Jeans   1920
5            Men Skinny Fit Jeans   1734
6  Men Slim Fit Stretchable Jeans   1699
7          Men 511 Slim Fit Jeans   1696
8          Men Straight Fit Jeans   1681
9                      Men Jeans   1342
```

[23]:
```
# Visualization of Top 10 Products "Pants and Trousers" Purchase by Consumers
plt.figure(figsize=(8,6))
plt.pie(Pants['count'][:10], labels = Pants['pants_description'][:10], autopct = '%.0f%%')
plt.show()
```

Pie Chart of Top 10 Products Purchase By Consumers



Visualization Findings

- Men Slim Fit Jeans enjoy a higher market share (30%) than any other jeans type.
- The category of Slim Fit Jeans has a considerable market share as well.
- The market shares for Mid-Rise Slim Fit Jeans and Relaxed Fit Jeans are identical at 12%.
- Anti Fit Jeans also enjoy a higher market share (8%) among all jean types.
- The market shares for Regular Fit Mid-Rise Jeans and Skinny Fit Jeans are equal at 7%.
- Stretchable Jeans, 511 Slim Fit Jeans, and Straight Fit Jeans all have market shares of about 6%.
- The market share for Men Jeans is the lowest (5%) among all jean types.

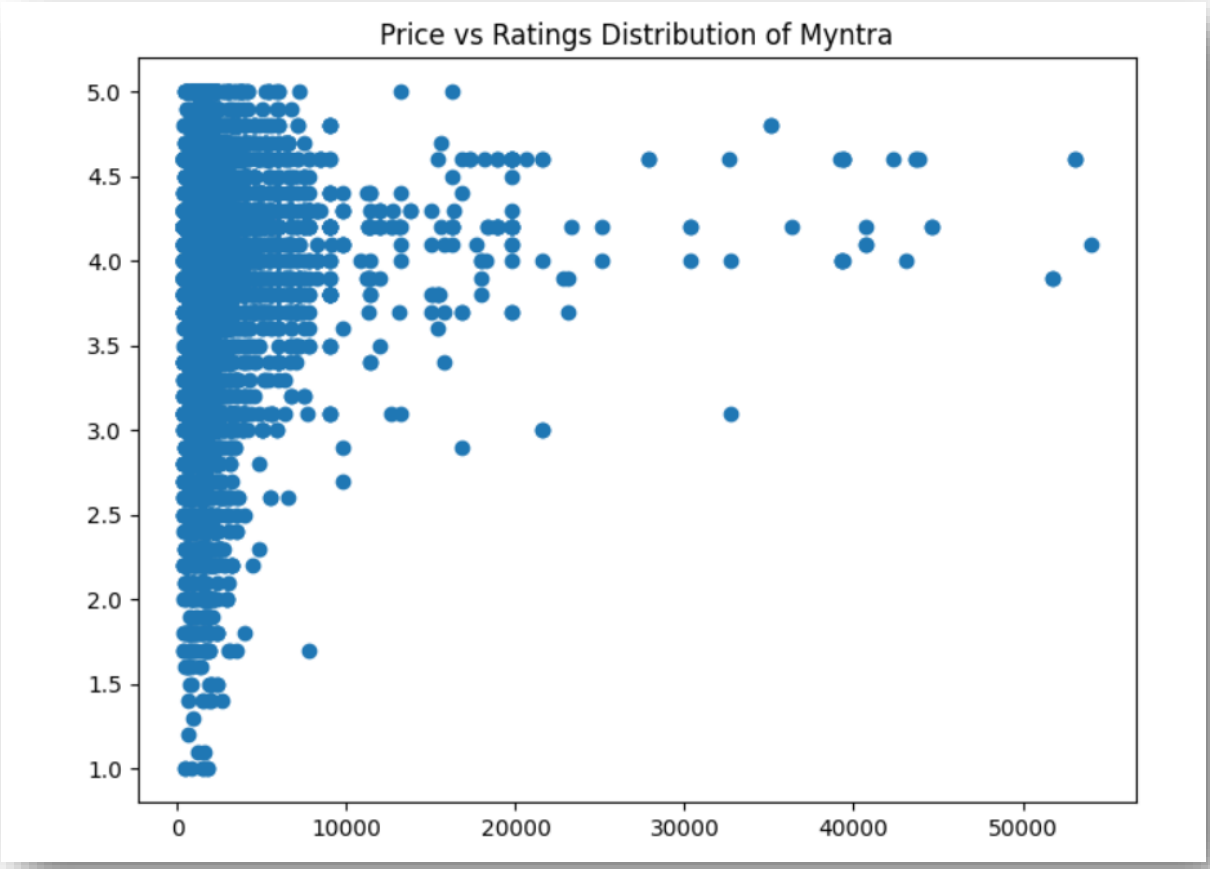
Bivariate Analysis and Multivariate Analysis

Bivariate analysis is defined as the analysis of the association between two variables, where one variable is influenced by or associated with another.

It can be conducted using software such as Pandas, Numpy, Matplotlib, and Seaborn in Python programming.

Scatterplot of Price vs Ratings Distribution of Myntra Dataset

```
# Price vs Ratings Distribution of Myntra Dataset
plt.figure(figsize=(8,6))
plt.scatter(x=df['price'], y=df['ratings'])
plt.title("Price vs Ratings Distribution of Myntra")
plt.show()
```



Visualization Findings

- The concentration of most products is in the lower price category, with scores between 3.5 and 4.5.
- No significant association exists between price and score.
- Expensive products are not automatically associated with high scores.
- Most scores are concentrated above 3.5, which suggests that there are more positive responses than negative ones.
- Some low scores can be seen in the lower price category.
- The number of expensive products is low, and they are spread out at similar score ranges.

Brand Wise Analysis of the Products

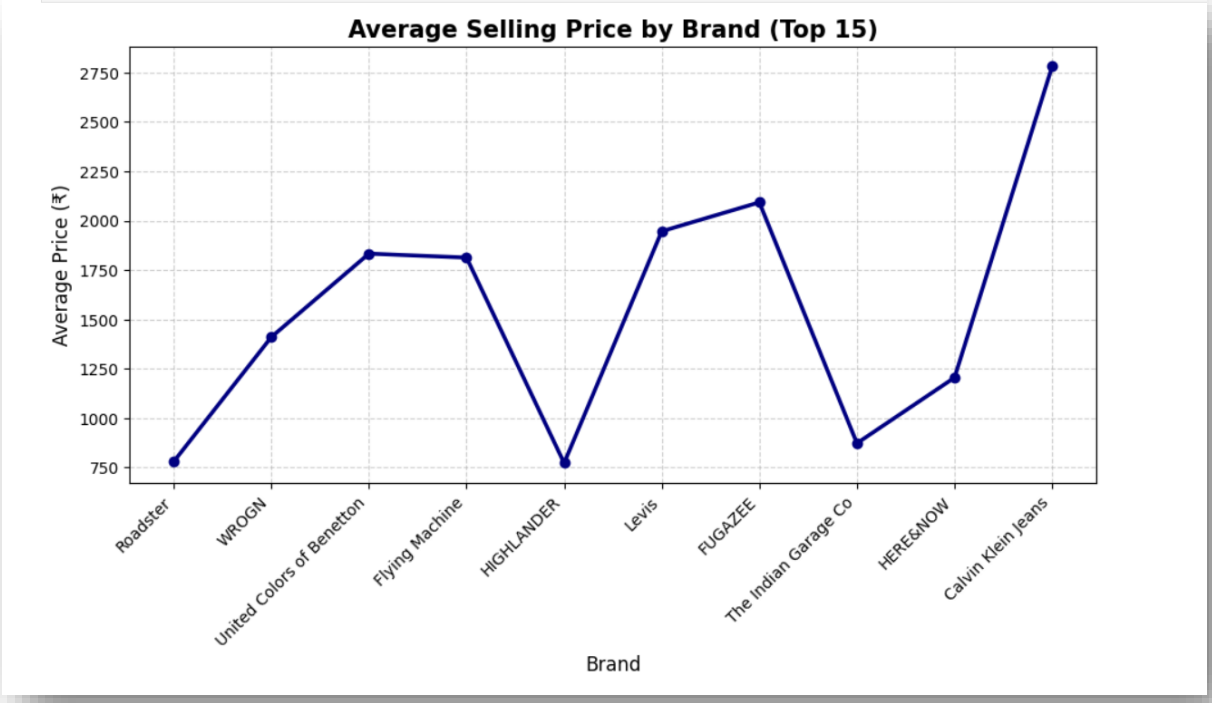
```
[29]: # Brand-wise Analysis
brand_analysis = df.groupby('brand_name').agg({
    'price': 'mean',
    'ratings': 'mean',
    'number_of_ratings': 'sum'
}).round(2).sort_values('price', ascending = False)
print(brand_analysis.head(10))
```

```
                         price  ratings  number_of_ratings
brand_name
Jacob Cohen            33142.50     4.08              512.0
Tramarossa             22345.05     4.30              752.0
Karl Lagerfeld         18487.50     4.45               79.0
Just Cavalli           18149.00     4.60                5.0
Versace Jeans Couture  17210.12     4.01              490.0
7 For All Mankind      16223.54     4.23              517.0
HACKETT LONDON         14220.50     4.10              302.0
True Religion          11153.64     4.09             7363.0
Polo Ralph Lauren      10601.70     3.99              644.0
TRAVIS MATHEW           8243.00     4.02              277.0
```

```
[32]: # Sort by number of ratings (most popular brands)
brand_analysis = brand_analysis.sort_values('number_of_ratings', ascending=False)
```

Line Graph of Brand Wise Analysis of the Products

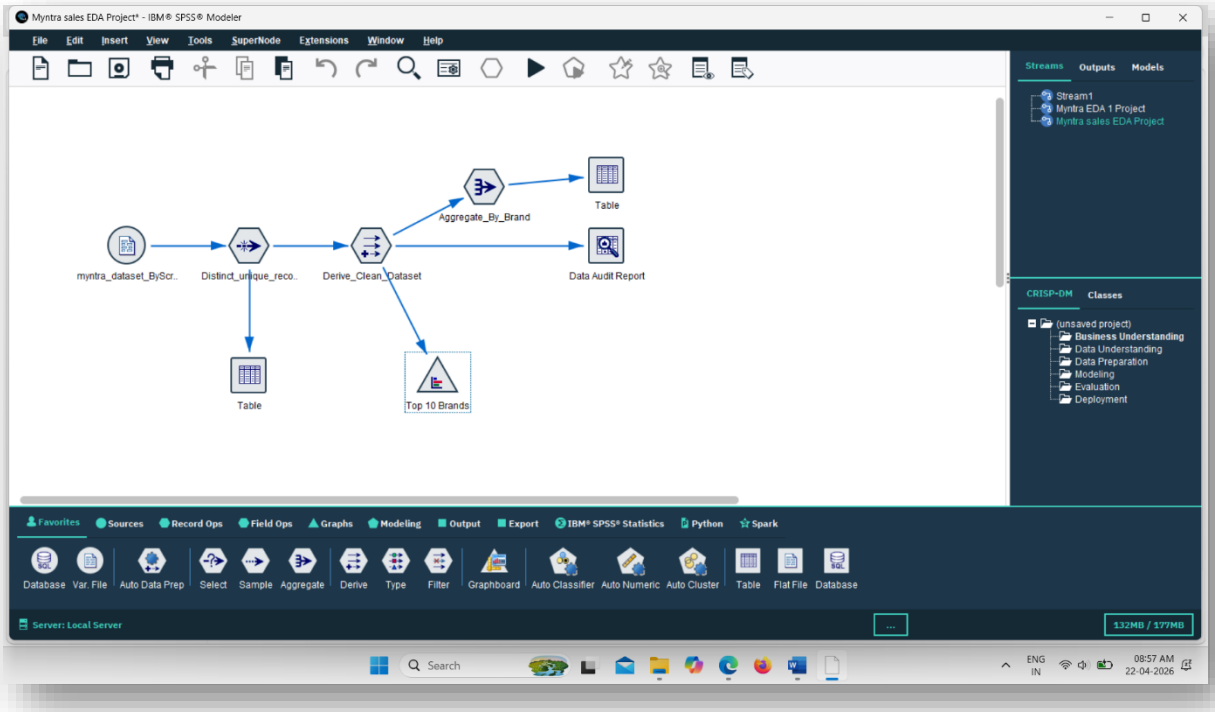
```
[60]: # Line Graph - Average Price of Top 15 Brands
plt.figure(figsize=(11, 5))
plt.plot(top_brands.index, top_brands['price'],
         marker='o', color='navy', linewidth=2.5, markersize=6)

plt.title('Average Selling Price by Brand (Top 15)', fontsize=15, fontweight='bold')
plt.xlabel('Brand', fontsize=12)
plt.ylabel('Average Price (₹)', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(True, linestyle='--', alpha=0.6)

plt.savefig('avg_price_line_graph.png', dpi=300, bbox_inches='tight')
plt.show()
```

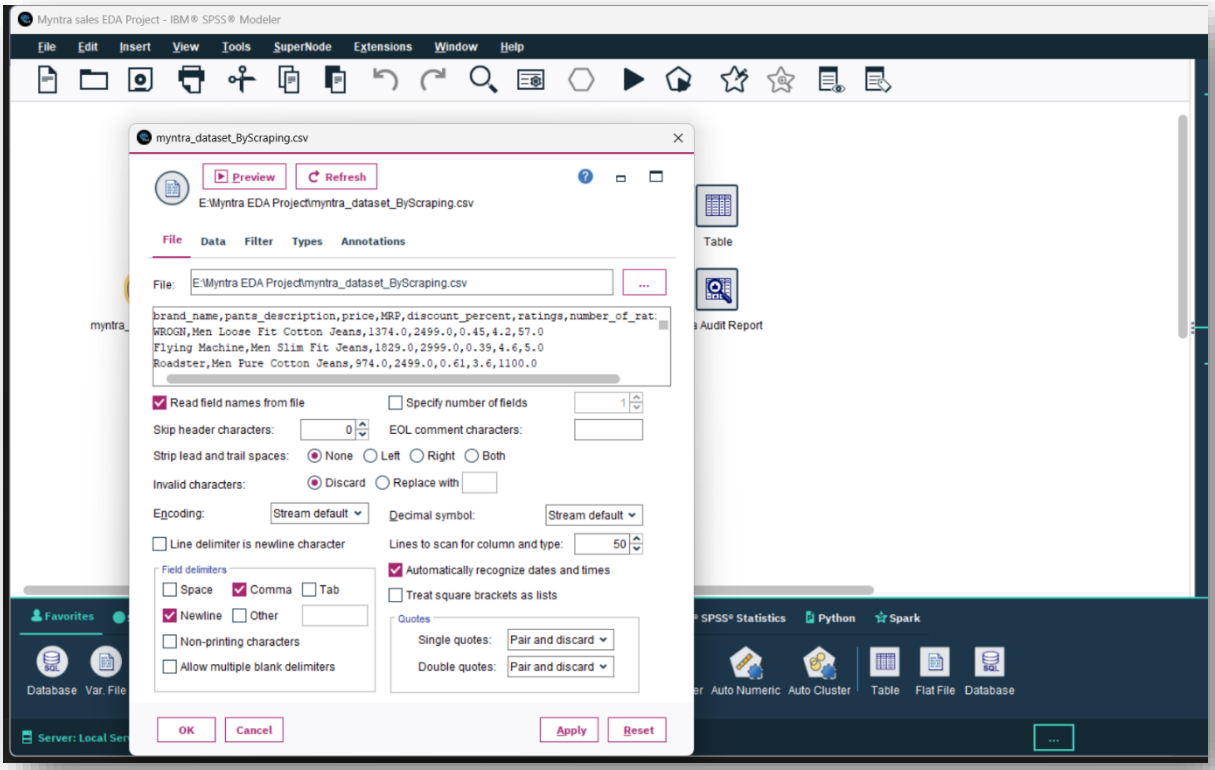


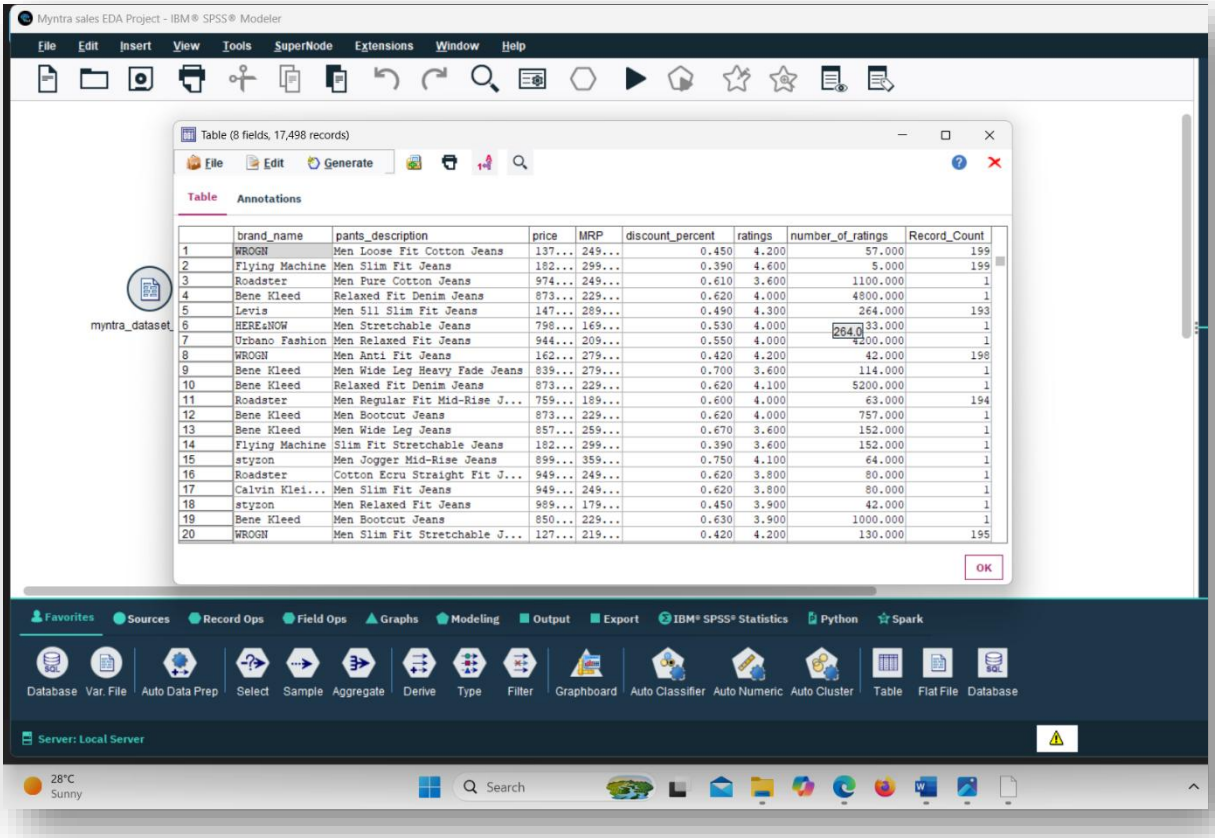
Visualization Findings

- Calvin Klein Jeans holds the position for the highest average selling price.
- FUGAZEE and Levis are placed under the category of expensive brands.
- United Colors of Benetton and Flying Machine have a higher average selling price.
- WROGN is placed in the mid-range price category.
- HERE&NOW and The Indian Garage Co provide comparatively low prices.
- Roadster and HIGHLANDER possess the lowest average selling prices.
- There is a visible distinction in the pricing of different categories of brands.
- This suggests significant brand-based price discrimination.

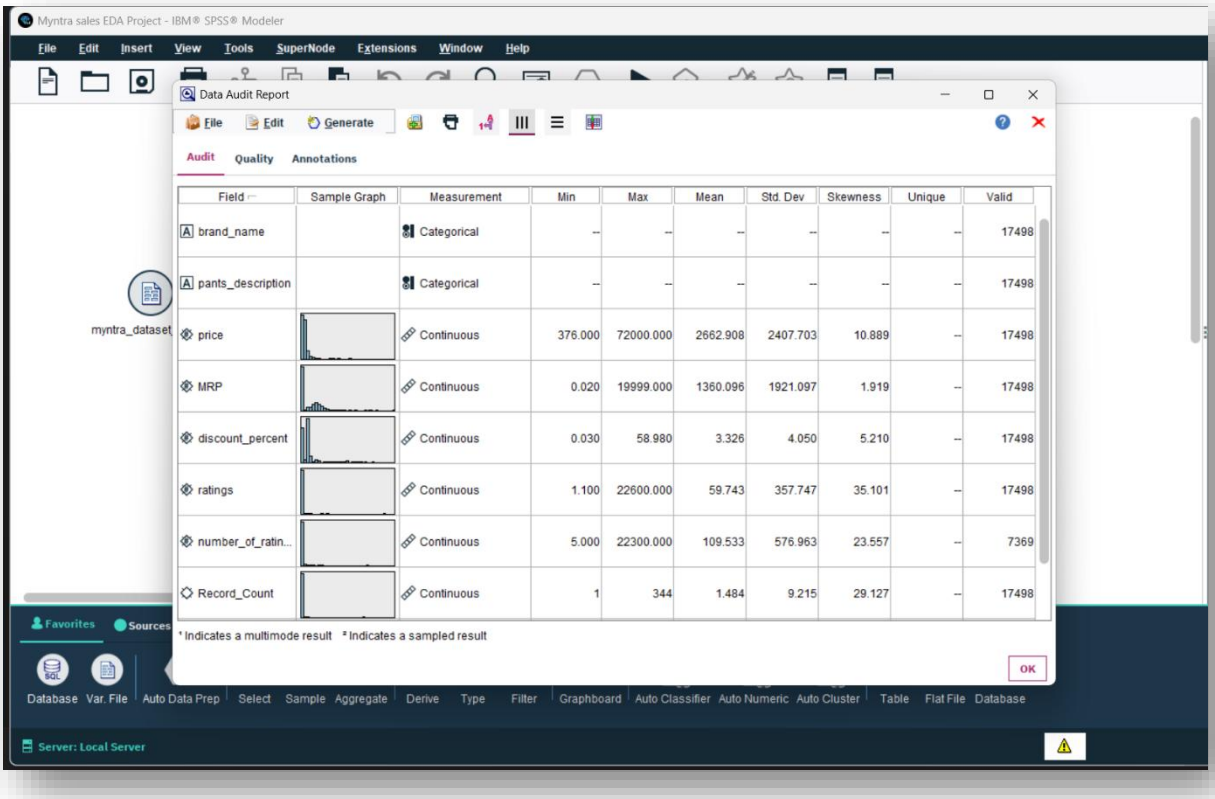
Step 5: Importing Data & Data Audit in SPSS Modeler







	brand_name	pants_description	price	MRP	discount_percent	ratings	number_of_ratings	Record_Count
1	WROGN	Men Loose Fit Cotton Jeans	137...	249...	0.450	4.200	57.000	199
2	Flying Machine	Men Slim Fit Jeans	182...	299...	0.390	4.600	5.000	199
3	Roadster	Men Pure Cotton Jeans	974...	249...	0.610	3.600	1100.000	1
4	Bene Kleed	Relaxed Fit Denim Jeans	873...	229...	0.620	4.000	4800.000	1
5	Levis	Men 511 Slim Fit Jeans	147...	289...	0.490	4.300	264.000	193
6	HERE&NOW	Men Stretchable Jeans	798...	169...	0.530	4.000	264.0 33.000	1
7	Urbano Fashion	Men Relaxed Fit Jeans	944...	209...	0.550	4.000	4200.000	1
8	WROGN	Men Anti Fit Jeans	162...	279...	0.420	4.200	42.000	198
9	Bene Kleed	Men Wide Leg Heavy Fade Jeans	839...	279...	0.700	3.600	114.000	1
10	Bene Kleed	Relaxed Fit Denim Jeans	873...	229...	0.620	4.100	5200.000	1
11	Roadster	Men Regular Fit Mid-Rise J...	759...	189...	0.600	4.000	63.000	194
12	Bene Kleed	Men Bootcut Jeans	873...	229...	0.620	4.000	757.000	1
13	Bene Kleed	Men Wide Leg Jeans	857...	259...	0.670	3.600	152.000	1
14	Flying Machine	Slim Fit Stretchable Jeans	182...	299...	0.390	3.600	152.000	1
15	styzon	Men Jogger Mid-Rise Jeans	899...	359...	0.750	4.100	64.000	1
16	Roadster	Cotton Ecru Straight Fit J...	949...	249...	0.620	3.800	80.000	1
17	Calvin Klei...	Men Slim Fit Jeans	949...	249...	0.620	3.800	80.000	1
18	styzon	Men Relaxed Fit Jeans	989...	179...	0.450	3.900	42.000	1
19	Bene Kleed	Men Bootcut Jeans	850...	229...	0.630	3.900	1000.000	1
20	WROGN	Men Slim Fit Stretchable J...	127...	219...	0.420	4.200	130.000	195



Field	Sample Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
brand_name		Categorical	--	--	--	--	--	--	17498
pants_description		Categorical	--	--	--	--	--	--	17498
price		Continuous	376.000	72000.000	2662.908	2407.703	10.889	--	17498
MRP		Continuous	0.020	19999.000	1360.096	1921.097	1.919	--	17498
discount_percent		Continuous	0.030	58.980	3.326	4.050	5.210	--	17498
ratings		Continuous	1.100	22600.000	59.743	357.747	35.101	--	17498
number_of_ratin...		Continuous	5.000	22300.000	109.533	576.963	23.557	--	7369
Record_Count		Continuous	1	344	1.484	9.215	29.127	--	17498

[1] Indicates a multimode result [2] Indicates a sampled result

The clean CSV file was loaded into SPSS Modeler through the use of the Source -> Var. File node. A Data Audit node was then added in order to automatically create a data quality report showing summary statistics, missing value check (shows zero missing values), and outliers (price, discount_percent).

Key Observations

- The dataset has 17,047 duplicate rows (same product listed multiple times due to scraping). We use Distinct Node to keep only unique records.
- Removing Duplicate Records using Distinct Node. The scraped dataset contained 17,047 duplicate rows. The Distinct node was applied on all fields to retain only unique product listings. After processing, the dataset was reduced to 34,073 unique records.
- The discount_percent field contained unrealistic values (maximum 64.0). A Derive node was used with the formula: if(discount_percent > 1, discount_percent / 100, discount_percent) This created a new field clean_discount_percent with values between 0 and 1, suitable for analysis.

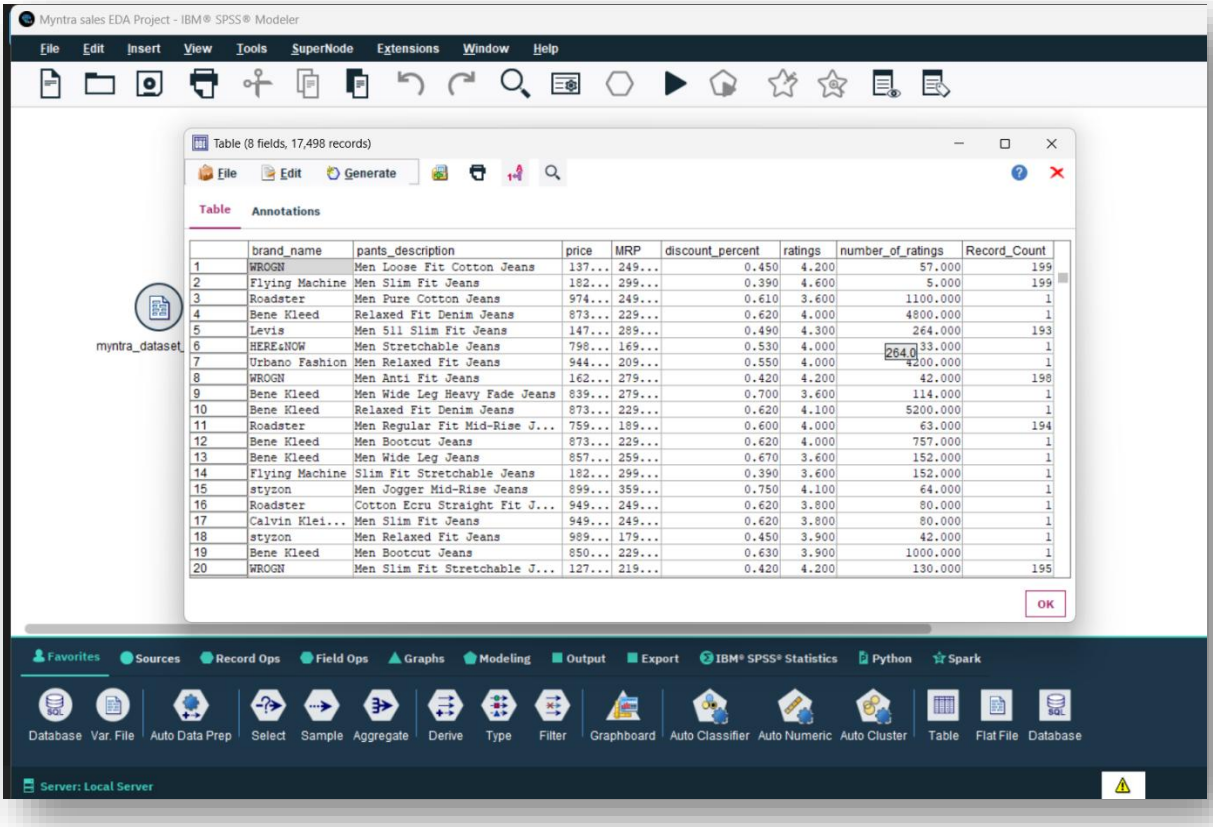
Step 6: Statistical Analysis & Graphing in SPSS Modeler

- In this we can use Statistics and Aggregate Node.
- Fo the Mean, Median, Standard Deviation, Maximum and Minimum value for Price, MRP, Ratings.
- In this we can use Graph node for the visualization of Price Distribution of Products, Discount Distribution, Price vs Ratings, MRP vs Ratings and Top 10 Brands by average ratings.

Key Observations

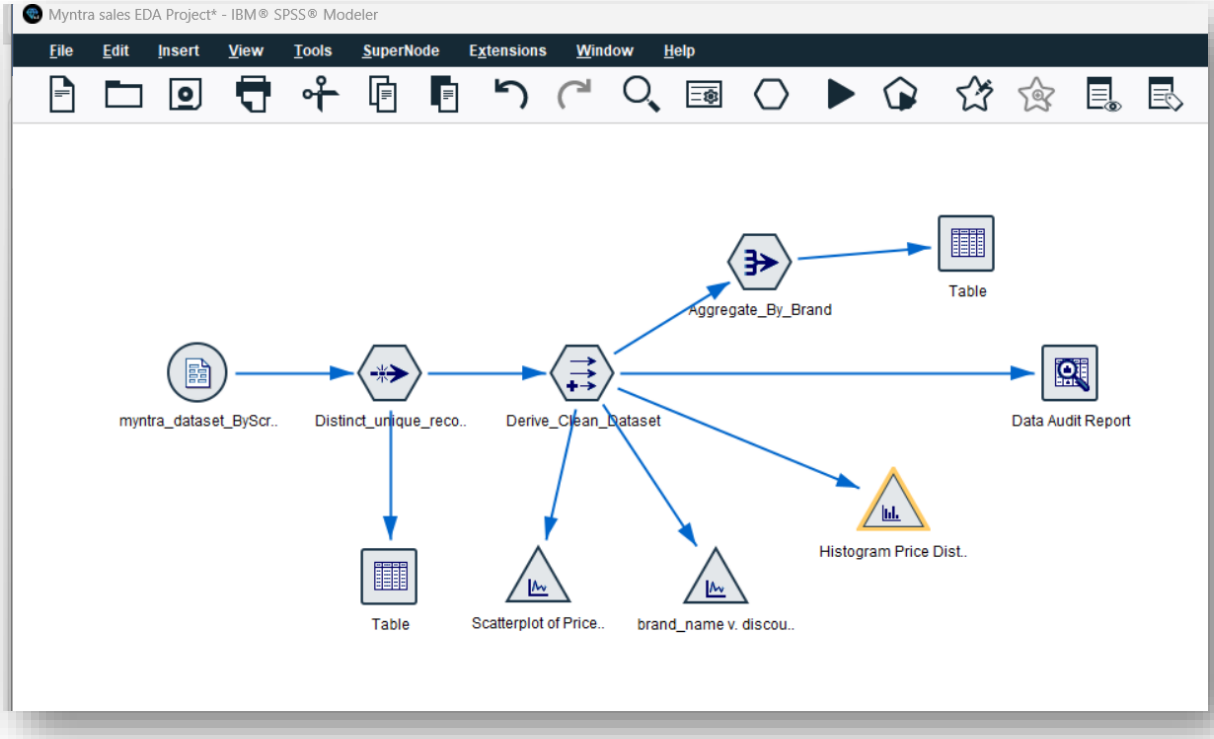
- An Aggregate node was used to group data by brand_name and compute mean price, mean ratings, total number of ratings, and average discount. This helped identify top-performing brands like WROGN, Flying Machine, and Roadster.

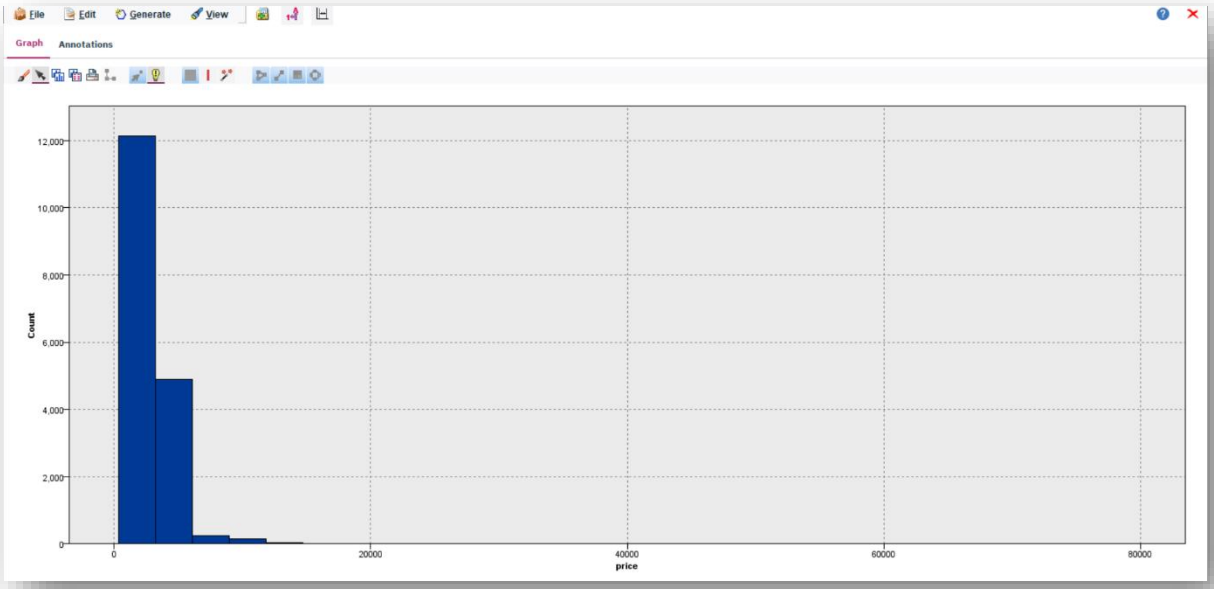
- Multiple Graph nodes were used to visualize data distribution. The Bar chart shows WROGN and Flying Machine as the most listed brands. The Histogram reveals that most jeans are priced between ₹500 – ₹2500.



	brand_name	pants_description	price	MRP	discount_percent	ratings	number_of_ratings	Record_Count
1	WROGN	Men Loose Fit Cotton Jeans	137...	249...	0.450	4.200	57.000	199
2	Flying Machine	Men Slim Fit Jeans	182...	299...	0.390	4.600	5.000	199
3	Roadster	Men Pure Cotton Jeans	974...	249...	0.610	3.600	1100.000	1
4	Bene Kleed	Relaxed Fit Denim Jeans	873...	229...	0.620	4.000	4800.000	1
5	Levis	Men 511 Slim Fit Jeans	147...	289...	0.490	4.300	264.000	193
6	HERE&NOW	Men Stretchable Jeans	798...	169...	0.530	4.000	264.0 33.000	1
7	Urbano Fashion	Men Relaxed Fit Jeans	944...	209...	0.550	4.000	4200.000	1
8	WROGN	Men Anti Fit Jeans	162...	279...	0.420	4.200	42.000	198
9	Bene Kleed	Men Wide Leg Heavy Fade Jeans	839...	279...	0.700	3.600	114.000	1
10	Bene Kleed	Relaxed Fit Denim Jeans	873...	229...	0.620	4.100	5200.000	1
11	Roadster	Men Regular Fit Mid-Rise J...	759...	189...	0.600	4.000	63.000	194
12	Bene Kleed	Men Bootcut Jeans	873...	229...	0.620	4.000	757.000	1
13	Bene Kleed	Men Wide Leg Jeans	857...	259...	0.670	3.600	152.000	1
14	Flying Machine	Slim Fit Stretchable Jeans	182...	299...	0.390	3.600	152.000	1
15	styzon	Men Jogger Mid-Rise Jeans	899...	359...	0.750	4.100	64.000	1
16	Roadster	Cotton Ecru Straight Fit J...	949...	249...	0.620	3.800	80.000	1
17	Calvin Klei...	Men Slim Fit Jeans	949...	249...	0.620	3.800	80.000	1
18	styzon	Men Relaxed Fit Jeans	989...	179...	0.450	3.900	42.000	1
19	Bene Kleed	Men Bootcut Jeans	850...	229...	0.630	3.900	1000.000	1
20	WROGN	Men Slim Fit Stretchable J...	127...	219...	0.420	4.200	130.000	195

Histogram of Price Distribution





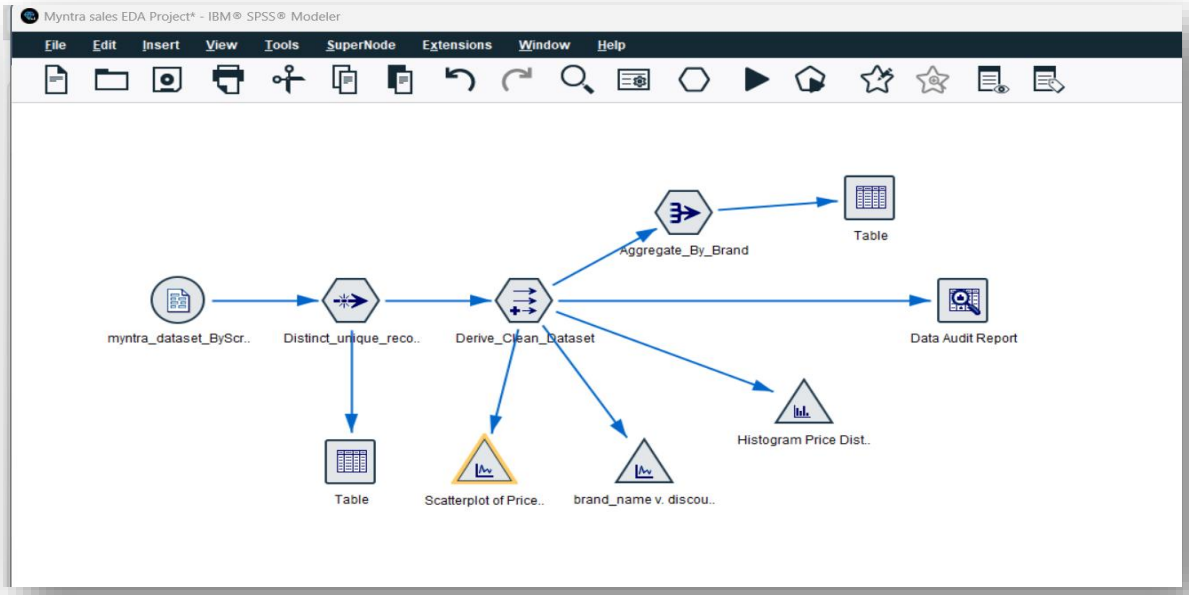
Interpretation:

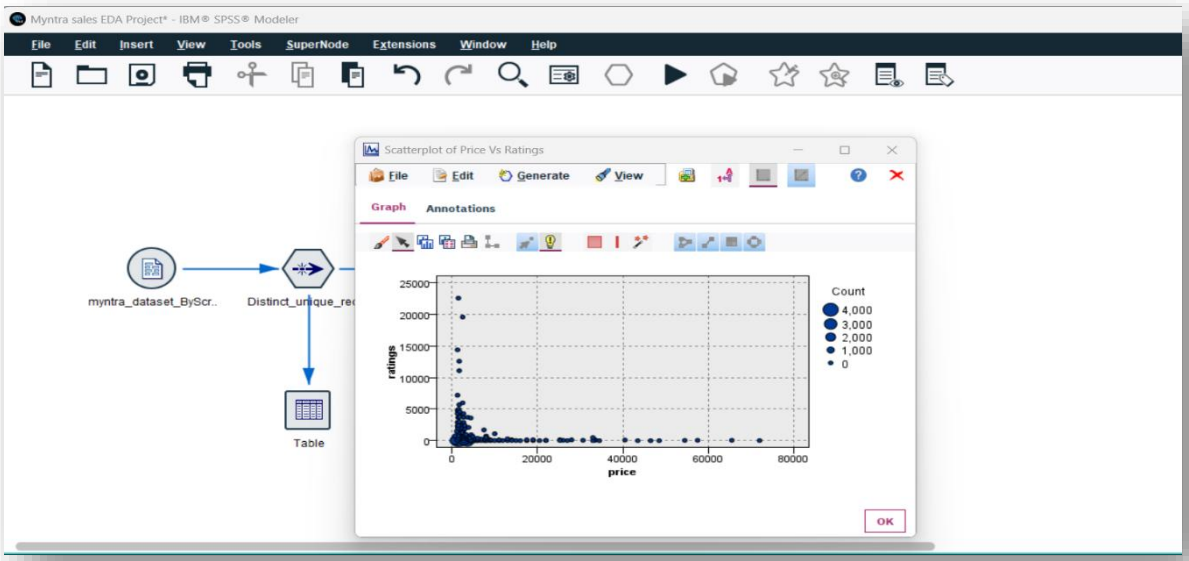
The histogram shows the distribution of the variable "price", which is highly positively (right-skewed). Most of the data points are highly concentrated towards the low end of the "price" scale, specifically between 0 to $5,000$, where the frequency count exceeds $12,000$ observations.

There is a steep and quick drop-off in terms of frequency counts beyond that point with another smaller cluster formed between $5,000$ and $10,000$.

Frequency becomes very small after the mark of $15,000$ while the upper limit of the x-axis reaches up to $80,000$. Therefore, there are some outliers of extreme high prices in comparison to the other observations within the dataset. In this case, we can assume that the mean value is going to be much higher than the median one due to outliers influence while the mode is obviously located at the lowest level of price.

Scatterplot of Price vs Ratings

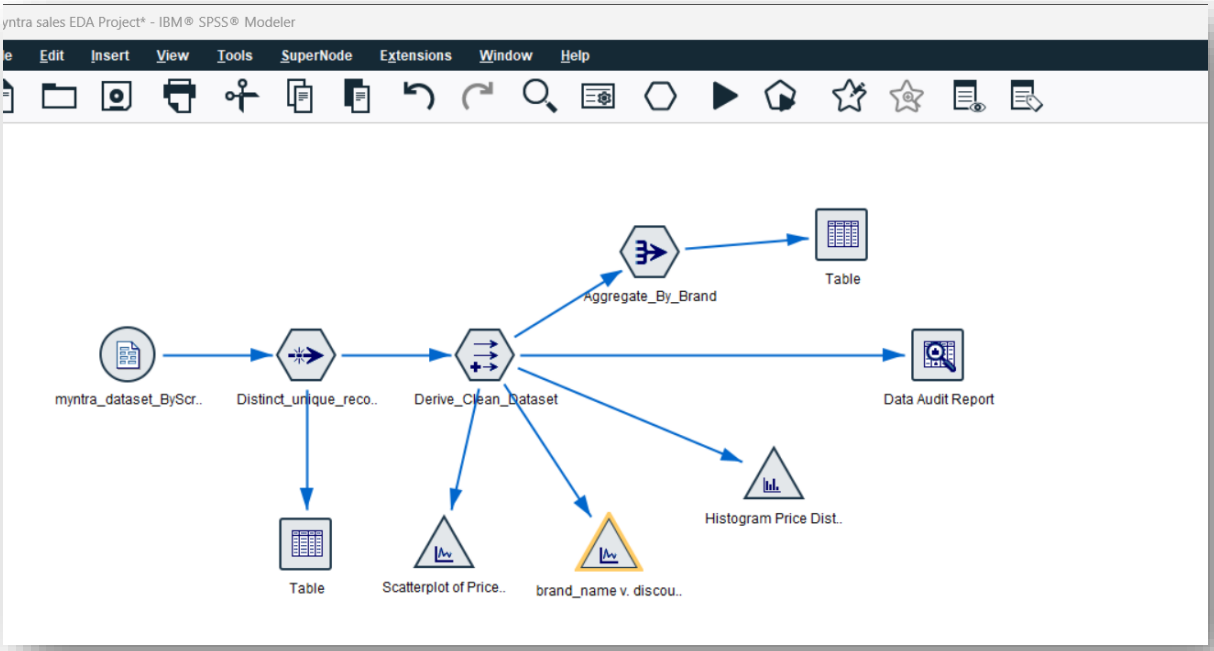


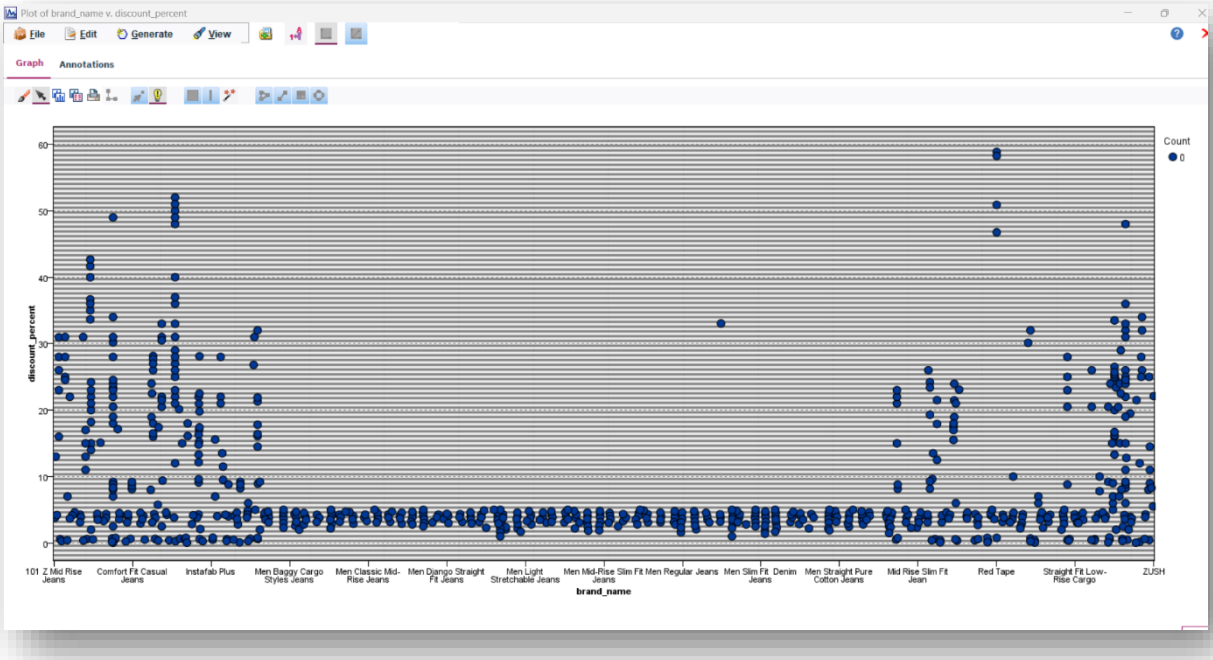


Interpretation:

This scatterplot shows the dependence of price on ratings (counts). As can be seen, there is a pronounced heavy bottom tail, meaning that a large number of points in the dataset cluster around smaller prices. In addition, we observe the inverse density tendency – when price rises, the count value drops sharply, with items costing more than $20,000$ practically having zero ratings. However, the most interesting and populated area seems to be the region of the plot in which the price range falls within 0-$5,000$. In this area, there are also numerous outliers whose counts exceed $10,000$, sometimes even reaching $20,000$, implying that only cheap items get much interaction. On the other hand, the points corresponding to high-priced items scattered along the x-axis up to $80,000$ are nearly devoid of any ratings at all.

Scatterplot of brands vs discounts

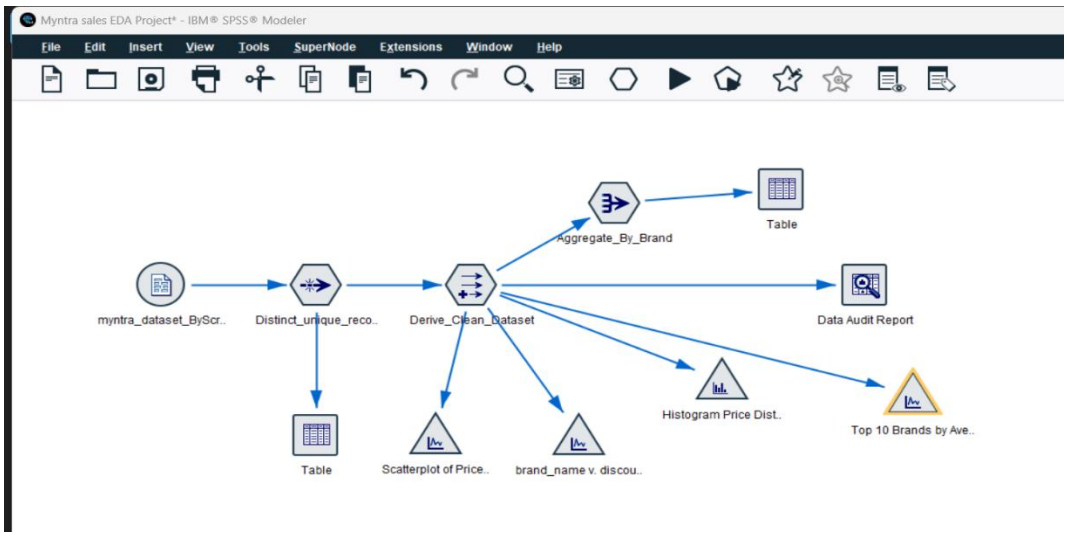


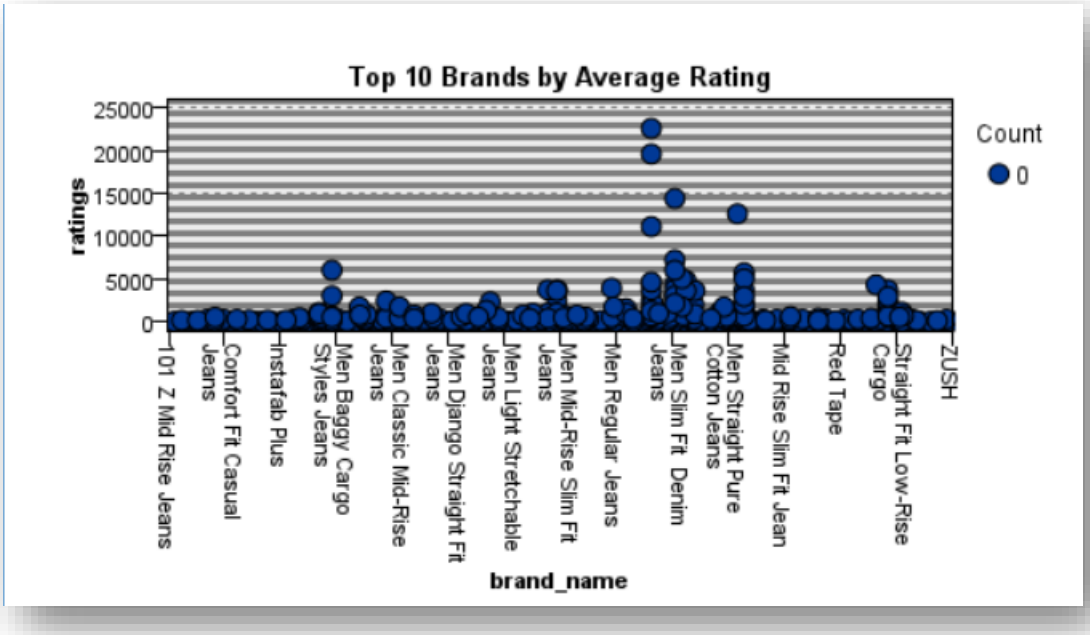


Interpretation:

The scatter plot shows that there is a great clustering of the discount percentage values in the range of $0% - $5%. Thus, one can conclude that most products from almost all brands get very small or even zero percent discounts. Nevertheless, some brands such as Insta fab Plus and Red Tape demonstrate a relatively great variability of discounts with a number of products being discounted by 40% up to 60%. At the same time, it should be said that while many brands from the middle part have a consistent approach to setting low discounts, brands located near the end of the x-axis are rather inconsistent in doing so.

Top 10 Brands by Average Ratings





Interpretation:

The scatterplot, titled "Top 10 Brands by Average Rating," visualizes the distribution of rating counts across various brand categories. While most brands show a consistent, low volume of ratings clustered near the baseline, **Men Slim Fit Denim Jeans** stands out as a significant outlier with multiple data points reaching between $10,000$ and $25,000$ ratings. This indicates that this specific category enjoys a much higher level of consumer engagement or sales volume compared to others like **Insta fab Plus** or **Red Tape**, which remain relatively flat. Other categories, such as **Men Baggy Cargo Styles Jeans** and **Men Straight Pure Cotton Jeans**, show moderate spikes in engagement, but the overall data suggests that market interaction is highly concentrated in specific "Slim Fit" and "Denim" segments.

Step 7: Deriving Meaningful Insights (Both Tools)

1. **Pricing Strategy:** Jeans are generally sold at a significant discount from MRP; discount_percent figures seem to show anomalies for some products (data entry issues).
2. **Performance of Brands:** Roadster, WROGN, and Flying Machine lead in sales and engagement. Premium brands (Levi's, Calvin Klein) enjoy high prices and ratings.
3. **Consumer Behaviour:** Low correlation between price and rating means that consumers place more importance on fit/description than price. Large number_of_ratings for low-cost brands shows mass-market appeal.

4. **Quality of Data:** 17,047 duplicate entries have been deleted using Python; SPSS Data Audit has flagged possible outliers.

Conclusion

EDA has been quite successful in identifying patterns in prices, dominant brands, and consumer preferences regarding the Myntra men's jeans category. Python has been very helpful in scripting along with providing customized visualizations, while SPSS Modeler has been used for fast and interactive visualization of data.

References

- Dataset: myntra_dataset_ByScraping.csv (Kaggle) •
- Tools: Python 3 + SPSS Modeler